```
from google.colab import files
uploaded = files.upload()   # yaha apni image select karo
image_path = list(uploaded.keys())[0]
```

Choose Files No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

Saving WhatsApp Image 2026-07-23 at 16.43.39 (3).jpeg to WhatsApp Image 2026-07-23 at 16.43.39 (3) (1).jpeg

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

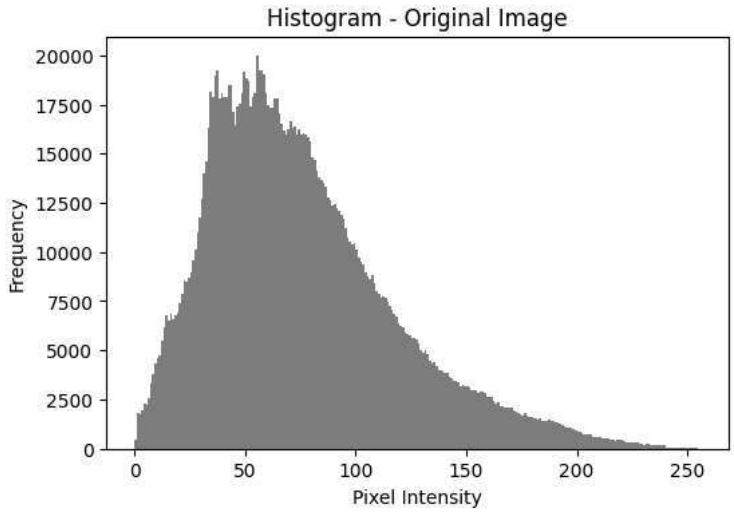
img = cv2.imread(image_path)
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

plt.figure(figsize=(5,5))
plt.imshow(gray, cmap='gray')
plt.title("Original Grayscale Image")
plt.axis('off')
plt.show()
```




```
import cv2
import numpy as np
import matplotlib.pyplot as plt

plt.figure(figsize=(6,4))
plt.hist(gray.ravel(), bins=256, range=[0,256], color='gray')
plt.title("Histogram - Original Image")
plt.xlabel("Pixel Intensity")
plt.ylabel("Frequency")
plt.show()
```




```
import cv2
import numpy as np
import matplotlib.pyplot as plt
```

```
def contrast_stretch(img):
    min_val = np.min(img)
    max_val = np.max(img)
    stretched = (img - min_val) * (255.0 / (max_val - min_val))
    return stretched.astype(np.uint8)

stretched_img = contrast_stretch(gray)

plt.figure(figsize=(5,5))
plt.imshow(stretched_img, cmap='gray')
plt.title("Contrast Stretched Image")
plt.axis('off')
plt.show()
```

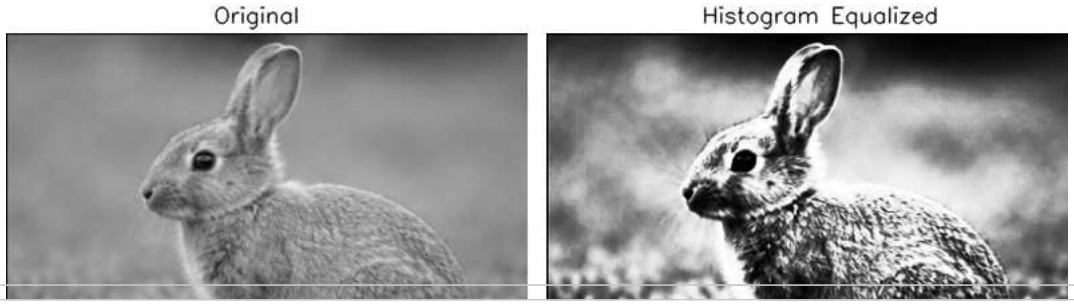



```
import cv2
import numpy as np
import matplotlib.pyplot as plt

equalized_img = cv2.equalizeHist(gray)

plt.figure(figsize=(10,5))
plt.subplot(1,2,1)
plt.imshow(gray, cmap='gray')
plt.title("Original")
plt.axis('off')

plt.subplot(1,2,2)
plt.imshow(equalized_img, cmap='gray')
plt.title("Histogram Equalized")
plt.axis('off')
plt.show()
```




```
import cv2
import numpy as np
import matplotlib.pyplot as plt

clahe = cv2.createCLAHE(clipLimit=2.0, tileGridSize=(8,8))
clahe_img = clahe.apply(gray)

plt.figure(figsize=(5,5))
plt.imshow(clahe_img, cmap='gray')
plt.title("CLAHE Enhanced Image")
plt.axis('off')
plt.show()
```




```
import cv2
import numpy as np
import matplotlib.pyplot as plt

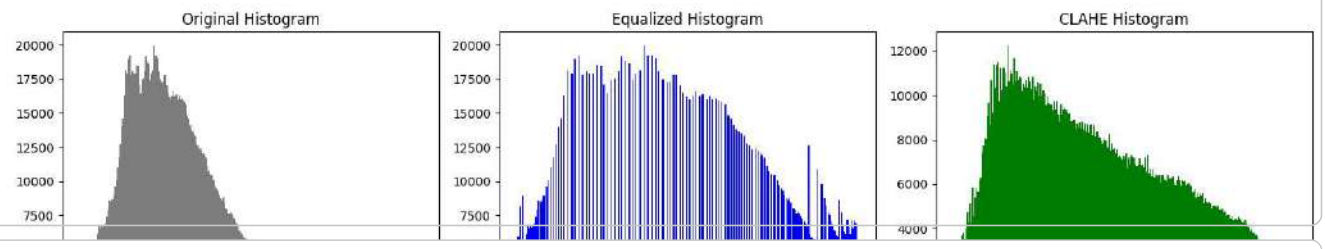
fig, axs = plt.subplots(1, 3, figsize=(15,4))

axs[0].hist(gray.ravel(), bins=256, range=[0,256], color='gray')
axs[0].set_title("Original Histogram")

axs[1].hist(equalized_img.ravel(), bins=256, range=[0,256], color='blue')
axs[1].set_title("Equalized Histogram")

axs[2].hist(clahe_img.ravel(), bins=256, range=[0,256], color='green')
axs[2].set_title("CLAHE Histogram")

plt.tight_layout()
plt.show()
```




```
import cv2
import numpy as np
import matplotlib.pyplot as plt

fig, axs = plt.subplots(1, 4, figsize=(18,5))

axs[0].imshow(gray, cmap='gray'); axs[0].set_title("Original"); axs[0].axis('off')
axs[1].imshow(stretched_img, cmap='gray'); axs[1].set_title("Contrast Stretched"); axs[1].axis('off')
axs[2].imshow(equalized_img, cmap='gray'); axs[2].set_title("Histogram Equalized"); axs[2].axis('off')
axs[3].imshow(clahe_img, cmap='gray'); axs[3].set_title("CLAHE"); axs[3].axis('off')

plt.tight_layout()
plt.show()
```

